
EE 511: Computer Vision

Course Code	EE 511
Course Name	Computer Vision
L-T-P-C	3-0-2-4
Prerequisites	Linear algebra, Probability and Statistics
Intended for	UG /MS / M. Tech. / PhD
Distribution	Elective: B. Tech III/IV year / MS / M. Tech./ PhD
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Essential mathematical tools:** Least squares, RANSAC, Eigen-analysis, PCA, SVD, clustering, gradient-based optimization methods. (4 Lectures)
- **Geometry, Camera models, Epipolar geometry, Stratified reconstruction, Applications:** large scale reconstruction, single-view metrology (8 Lectures)
- **Probabilistic graphical models:** MRF, CRF, Combinatorial optimization methods (5 Lectures)
- **Module IV:** Stereo disparity estimation, Optical flow (Lucas Kanade and Horn Schunk approaches, contemporary energy minimization methods) (5 Lectures)
- **Features detection and tracking:** Harris corner detector, KL tracking, SIFT, Overview of other contemporary descriptors. (5 Lectures)
- **Segmentation:** Low-level segmentation, energy minimization and clustering based methods, semantic segmentation (5 Lectures)
- **High level vision:** CNN overview, single image depth estimation, Flow-net, 3D scene understanding and segmentation. (6 Lectures)
- **Synthesis:** GAN overview, 3D shape synthesis, integrating viewpoint and texture, semantic image synthesis (4 Lectures)

Text Books

1. D. Forsyth and J. Ponce, **Computer vision- A modern approach**, 2nd edition, Pearson, 2012.

References

1. R. Hartley and A. Zisserman, **Multiple view geometry in computer vision**, 2nd edition, Cambridge university press, 2003
2. S. Prince, **Computer vision- Models, learning and inference**, Cambridge university press, 2012.
3. C. Bishop, **Pattern Recognition and Machine learning**, Springer, 2006.